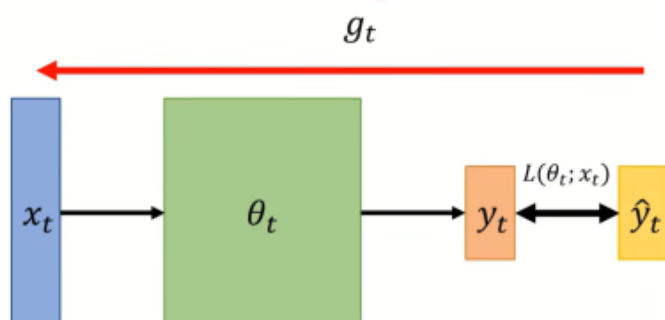


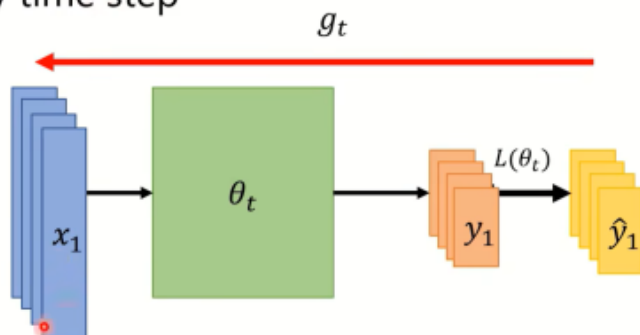
New Optimazation For Deep Learning

some Notations

- θ_t :在第t步的模型参数
 - $\nabla L(\theta_t)$ or g_t : θ_t 的梯度, 用来计算 θ_{t+1}
 - m_{t+1} :从第0步到第t步的动量, 用来计算 θ_{t+1}
- On-line : one pair of $(x_t, \hat{y}_t)$ at a time step



- Off-line : pour all $(x_t, \hat{y}_t)$ into the model at every time step



- On-line learning:一次处理一次一个x和y
- Off-line learning:一次处理所有的x和y

20年主流模型都在使用Adam和SGDM进行优化

- Adam : fast training, large generalization gap, unstable
- SGDM : stable, little generalization gap, better convergence(?)

An intuitive illustration for generalization gap

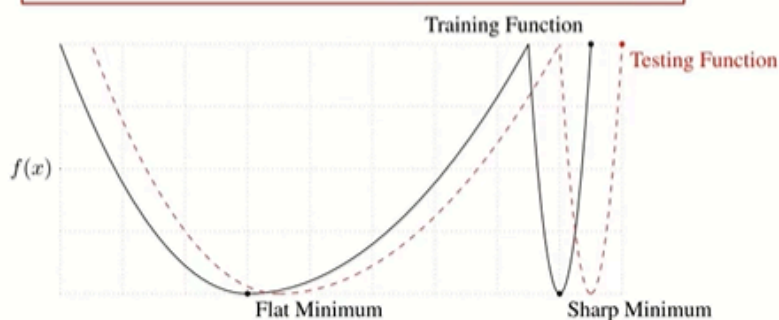


Figure 1: A Conceptual Sketch of Flat and Sharp Minima. The Y-axis indicates value of the loss function and the X-axis the variables (parameters)

Adam能更快的训练，泛化间隙大，但不稳定

SGDM更稳定，泛化间隙小，收敛效果好

SWATS

考虑将Adam和SGDM结合

一开始先用Adam，最后收敛的时候用SGDM

考虑优化Adam

AMSGrad

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t$$

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_{t-1}, \beta_1 = 0.$$

$$v_t = \beta_2 v_{t-1} + (1 - \beta_2) (g_{t-1})^2, \beta_2 = 0.999$$

The "memory" of v_t keeps roughly 1000 steps!!

In the final stage of training, most gradients are small and non-informative, while some mini-batches provide large informative gradient rarely.

time step	...	100000	100001	100002	100003	...	100999	101000
gradient		1	1	1	1		100000	1
movement		η	η	η	η		$10\sqrt{10}\eta$	$10^{-3.5}\eta$

当前面的 g 都很小的时候，在后面即使出现一个很大的 g ，对之后 g 的影响也会很小

- AMSGrad [Reddi, et al., ICLR'18]

$$\theta_t = \theta_{t-1} - \frac{\eta}{\sqrt{\hat{v}_t + \epsilon}} m_t$$

$$\hat{v}_t = \max(\hat{v}_{t-1}, v_t)$$

考虑记录并使用最大的 v_t ，会出现和Adagrad同样的问题，后期更新过慢

AdaBound

$$\theta_t = \theta_{t-1} - \text{Clip}\left(\frac{\eta}{\sqrt{\hat{v}_t + \epsilon}}\right) \hat{m}_t$$

$$\text{Clip}(x) = \text{Clip}\left(x, 0.1 - \frac{0.1}{(1 - \beta_2)t + 1}, 0.1 + \frac{0.1}{(1 - \beta_2)t}\right)$$

注： β_2 即Adam中当分母的 β_2

把learning rate做Clip，但Clip函数是经验公式，lower bound和upper bound已经固定，不算真正的Adaptive learning rate

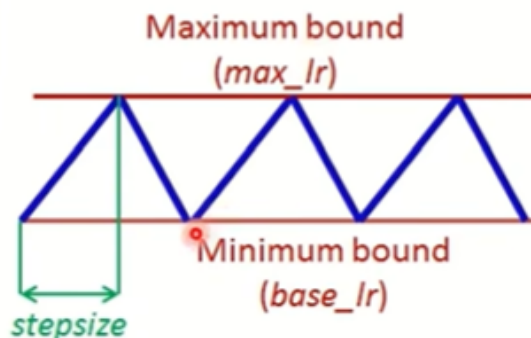
∴不好用

考虑优化SGDM

考虑将learning rate一会调大，一会调小

- Cyclical LR [Smith, WACV'17]

- learning rate : decide by LR range test
- stepsize : several epochs
- avoid local minimum by varying learning rate



- SGDR [Loshchilov, et al., ICLR'17]

